

Horizontal and vertical dilations



1

a 2π

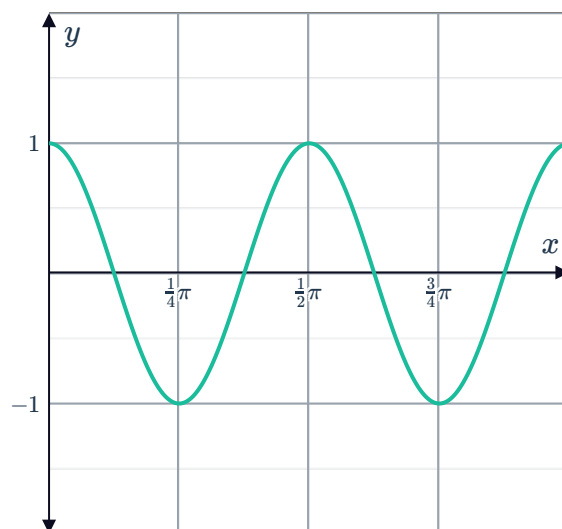
b

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$	$\frac{5\pi}{8}$	$\frac{3\pi}{4}$	$\frac{7\pi}{8}$	π
$g(x)$	1	0	-1	0	1	0	-1	0	1

c $\frac{1}{2}\pi$

d Horizontal dilation by a factor of $\frac{1}{4}$

e

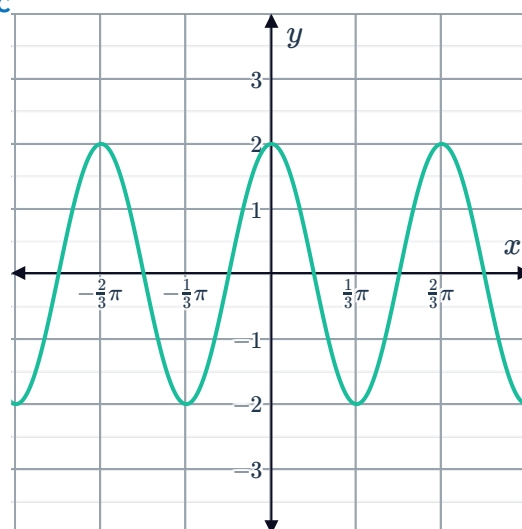


2

a 2

b $\frac{2\pi}{3}$

c



3

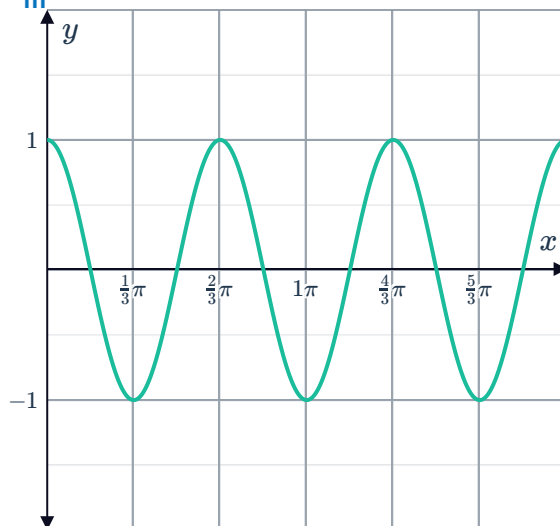
a

i $A = 1$

ii $P = \frac{2\pi}{3}$



iii

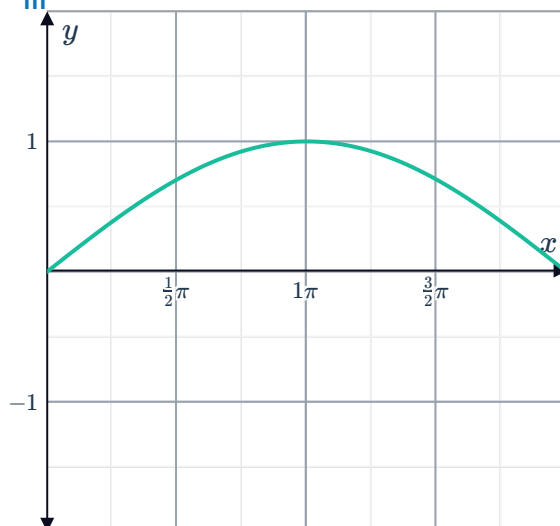


b

i $A = 1$

ii $P = 4\pi$

iii

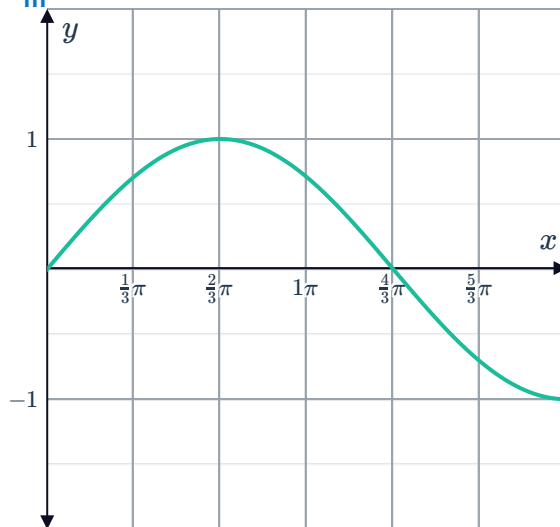


c

i $A = 1$

ii $P = \frac{8\pi}{3}$

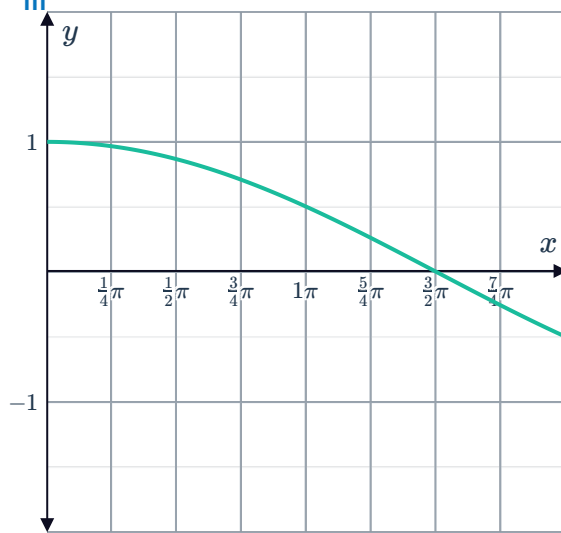
iii



d

i $A = 1$

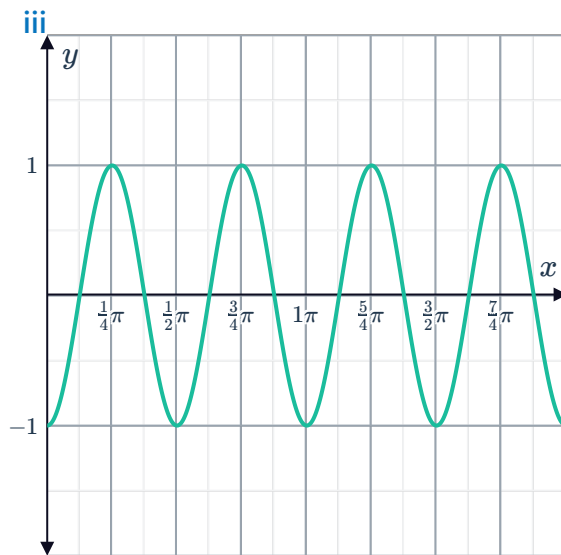
ii $P = 6\pi$



e

i $A = 1$

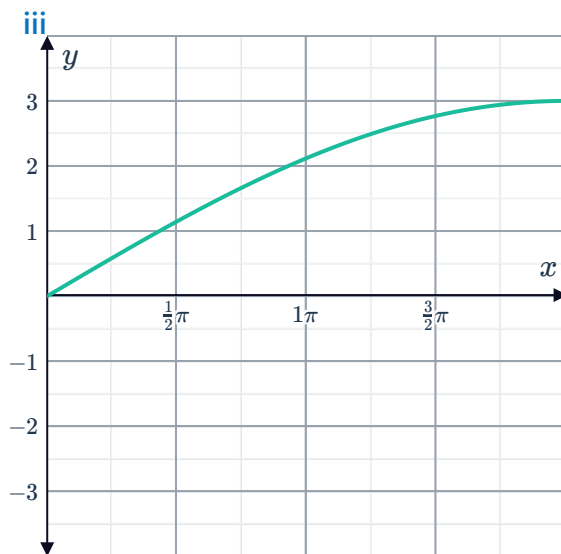
ii $P = \frac{\pi}{2}$



f

i $A = 3$

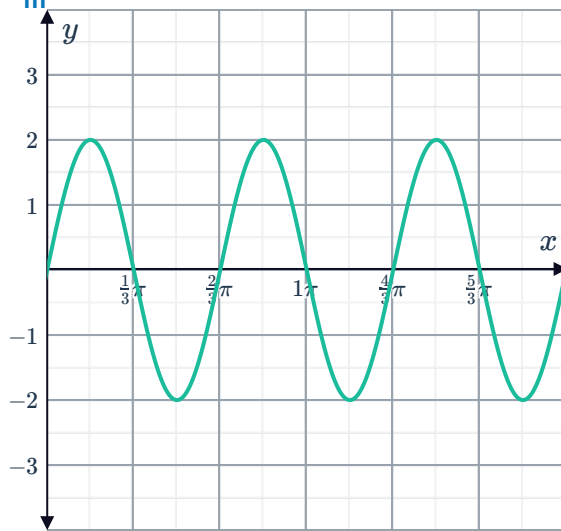
ii $P = 8\pi$



g

i $A = 2$

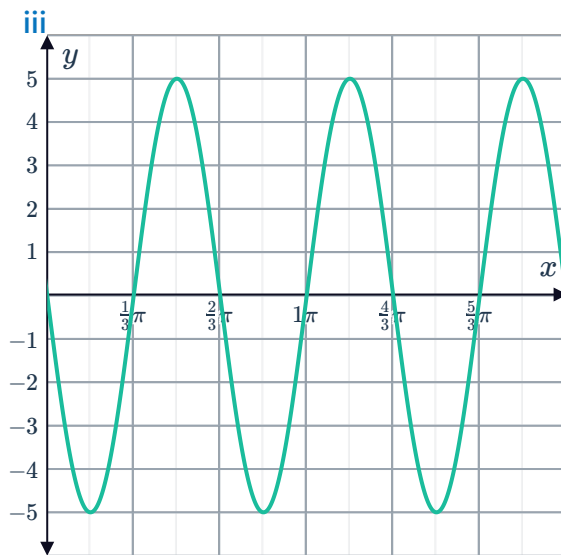
ii $P = \frac{2\pi}{3}$



h

i $A = 5$

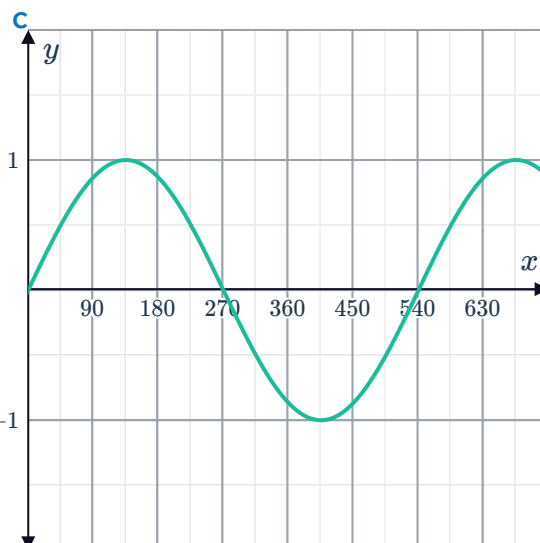
ii $P = \frac{2\pi}{3}$



4

a 1

b 540°



a	x	0	2π	4π	6π	8π		b	8π
	$\sin\left(\frac{x}{4}\right)$	0	1	0	-1	0			

6

a b $\frac{\pi}{b}$

Function	Period
$\tan x$	π
$\tan 2x$	$\frac{\pi}{2}$
$\tan 3x$	$\frac{\pi}{3}$
$\tan 4x$	$\frac{\pi}{4}$

c Shorter

7

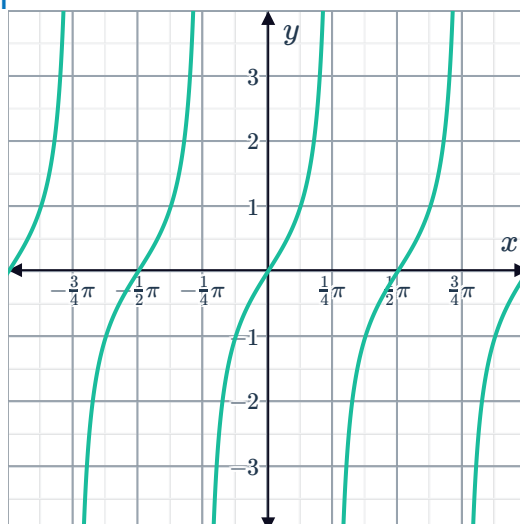
a $\frac{2\pi}{3}$ b 1 c $\frac{\pi}{2}$ d 4π
e 2 f $\frac{4\pi}{3}$ g $\frac{3\pi}{2}$ h 2

8

a Horizontal dilation by a factor of $\frac{1}{4}$. b 4

9

a i 0 ii $\frac{\pi}{2}$ iii $\frac{\pi}{2}$ iv $x = \frac{\pi}{4}$
v $x = -\frac{\pi}{4}$ vi



b

i 0

v $x = -\frac{\pi}{6}$

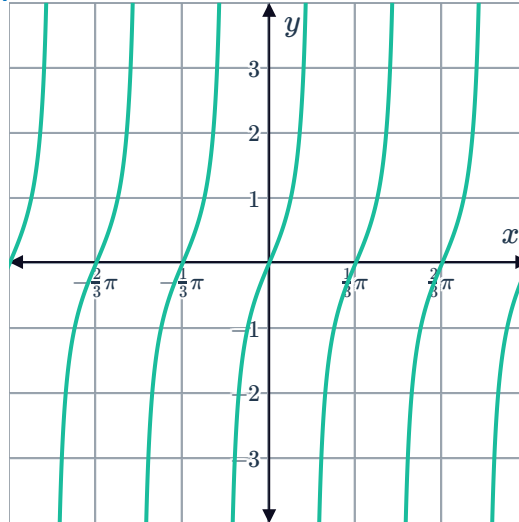
ii $\frac{\pi}{3}$



iii $\frac{\pi}{3}$

iv $x = \frac{\pi}{6}$

vi



c

i 0

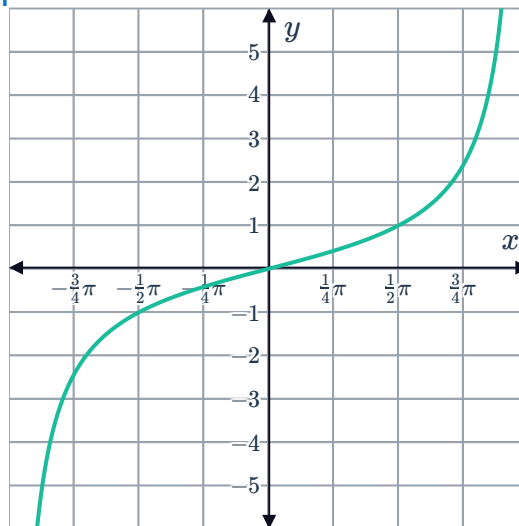
v $x = -\pi$

ii 2π

iii 2π

iv $x = \pi$

vi



d

i 0

v $x = -\frac{\pi}{8}$

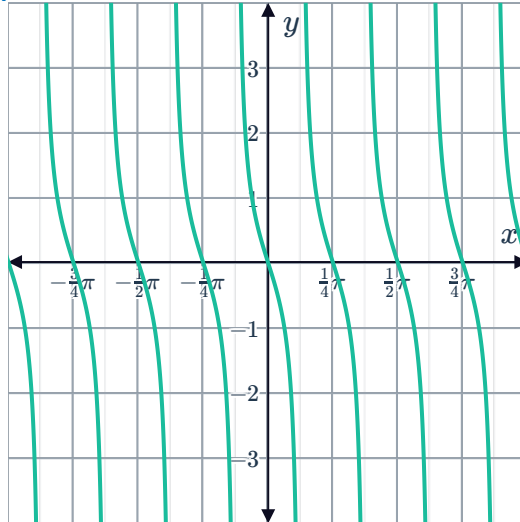
ii $\frac{\pi}{4}$



iii $\frac{\pi}{4}$

iv $x = \frac{\pi}{8}$

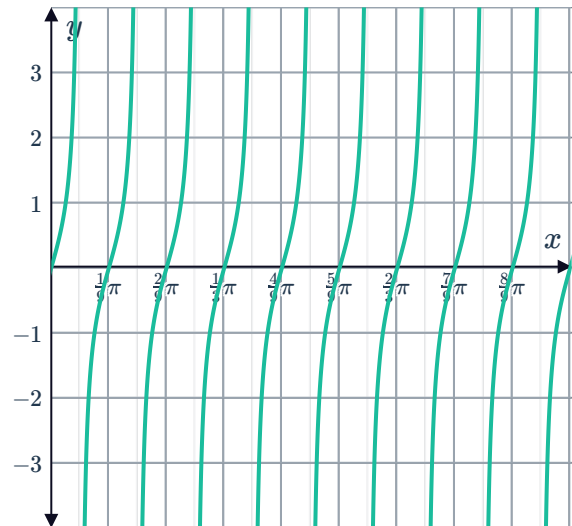
vi



10

a $\frac{\pi}{9}$

b



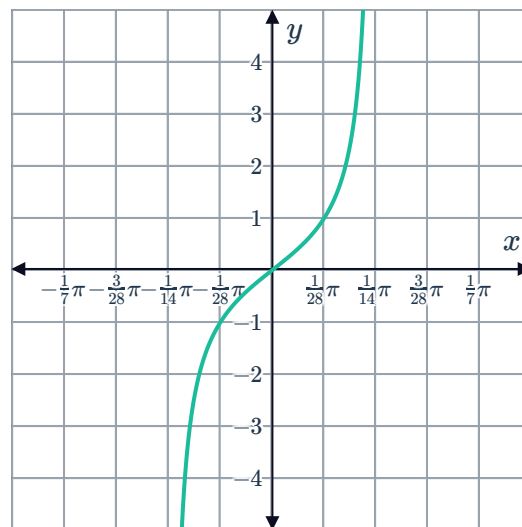
11



a

x	$-\frac{\pi}{28}$	0	$\frac{\pi}{28}$	$\frac{3\pi}{28}$	$\frac{\pi}{7}$	$\frac{5\pi}{28}$
y	-1	0	1	-1	0	1

b



12

a $\frac{53\pi}{24}$

b $\frac{\pi}{12}$

c $y = \tan 12x$

13 β is larger as $g(x)$ has a shorter period.

14

a

x	0	$\frac{3\pi}{2}$	3π	$\frac{9\pi}{2}$	6π
$\tan\left(\frac{x}{6}\right)$	0	1	undefined	-1	0

b 6π

d $x = 3\pi, x = 9\pi, x = 15\pi$

e -3π

Phase shifts

15

a 1

b -1

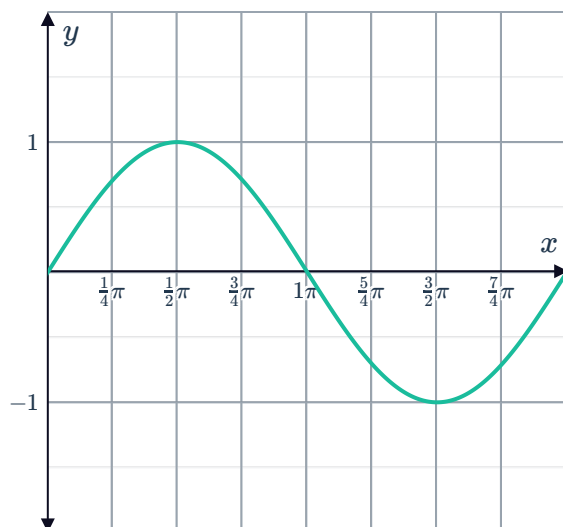
16

a

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$	1	0	-1	0	1
$g(x)$	0	1	0	-1	0

b Translated $\frac{\pi}{2}$ units to the right.

c



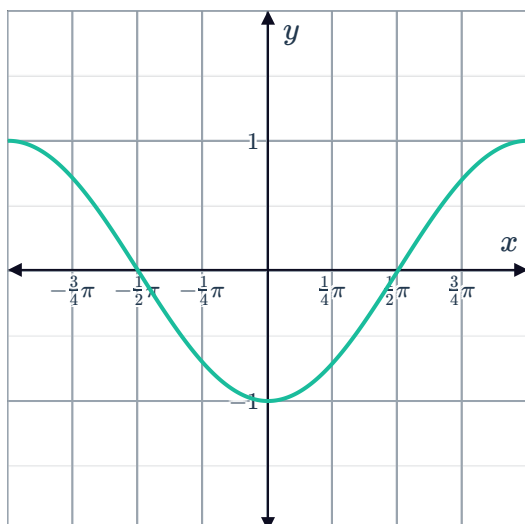
17

a

i 1

ii Translated $\frac{\pi}{2}$ units to the right.

iii



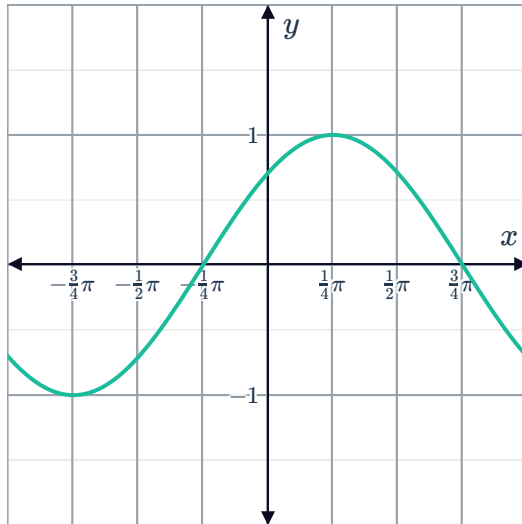
b

i 1



ii Translated $\frac{\pi}{4}$ units to the left.

iii



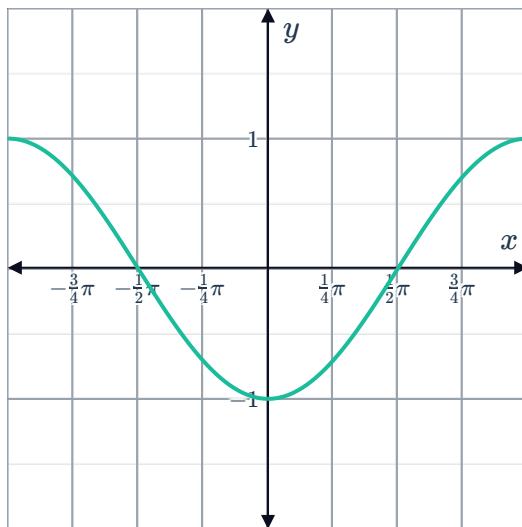
18

a

i 1

ii Translated π units to the left.

iii



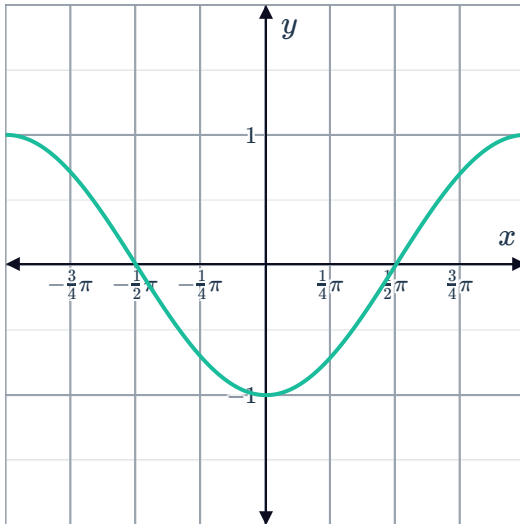
b

i 1



ii Translated π units to the right.

iii

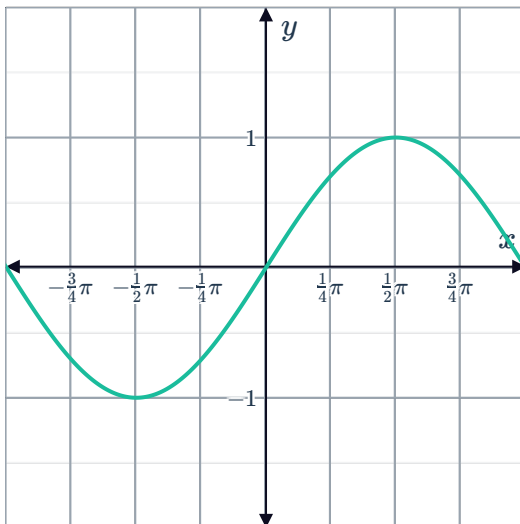


c

i 1

ii Translated $\frac{3\pi}{2}$ units to the left.

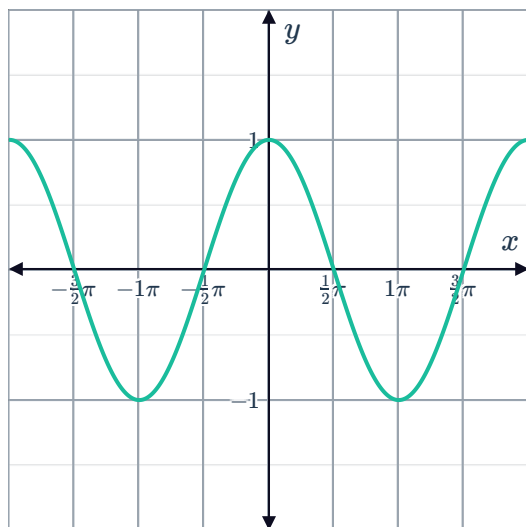
iii



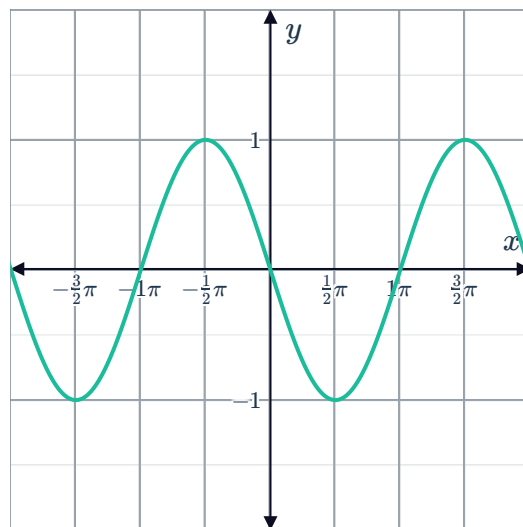
19 $c = -\frac{1}{2}\pi, \frac{3}{2}\pi$

20

a



b



21 True

22

a $\frac{\pi}{2}$ units to the left.

c 1.5 units to the right.

b $\frac{\pi}{5}$ units to the right.

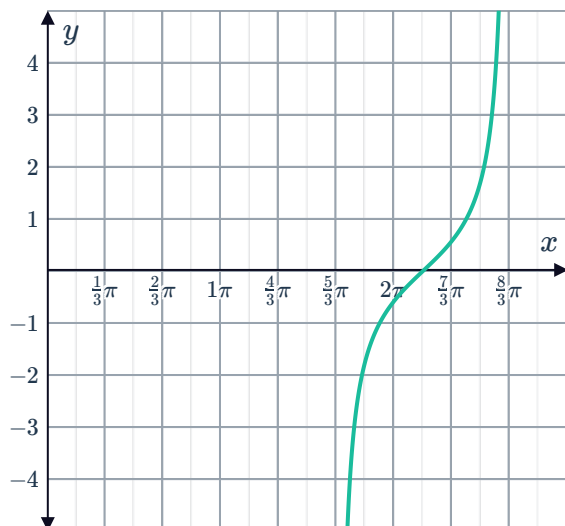
d $\frac{3\pi}{4}$ units to the left.

23

a $x = \frac{2\pi}{3}, x = \frac{5\pi}{3}, x = \frac{8\pi}{3}$

b $x = \frac{7\pi}{6}, \frac{13\pi}{6}$

c



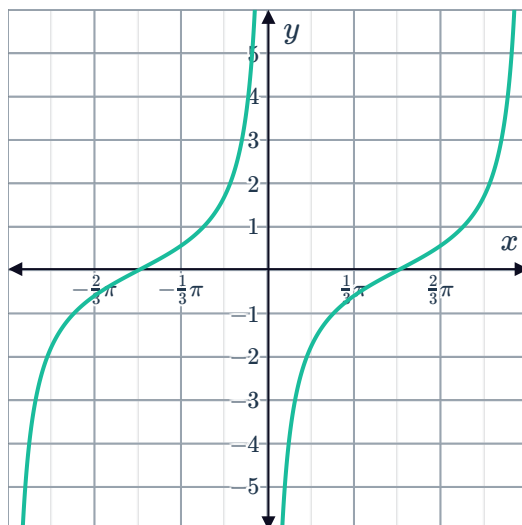
24



a

i There is no y -intercept.iii π v $x = 0$ ii π iv $x = \pi$

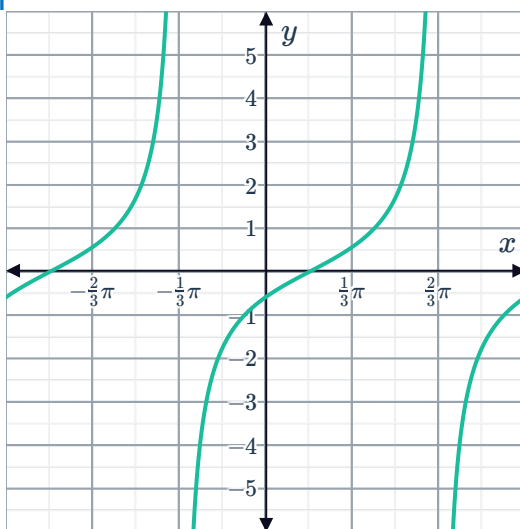
vi



b

i $-\frac{1}{\sqrt{3}}$ v $x = -\frac{\pi}{3}$ ii π iii π iv $x = \frac{2\pi}{3}$

vi



c

i $\sqrt{3}$

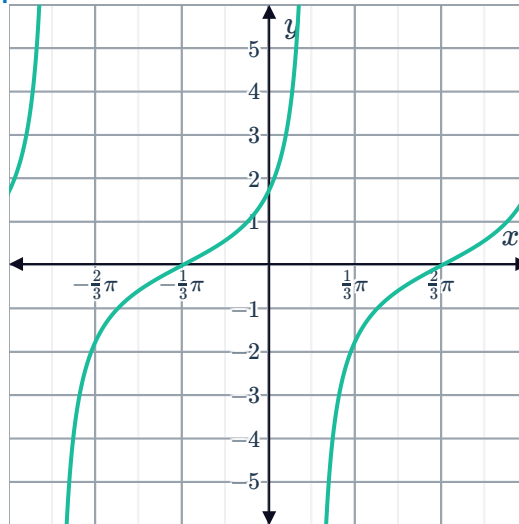
ii π



iv $x = \frac{\pi}{6}$

v $x = -\frac{5\pi}{6}$

vi



25

a $\frac{\pi}{7}$ units to the right

b $x = \frac{9\pi}{14}, x = \frac{23\pi}{14}, x = \frac{37\pi}{14}, x = \frac{51\pi}{14}$

Combined translations and dilations

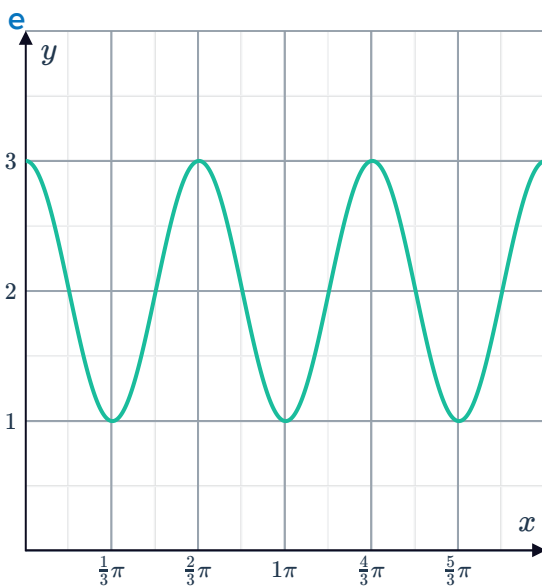
26

a $\frac{2\pi}{3}$

b 1

c 3

d 1



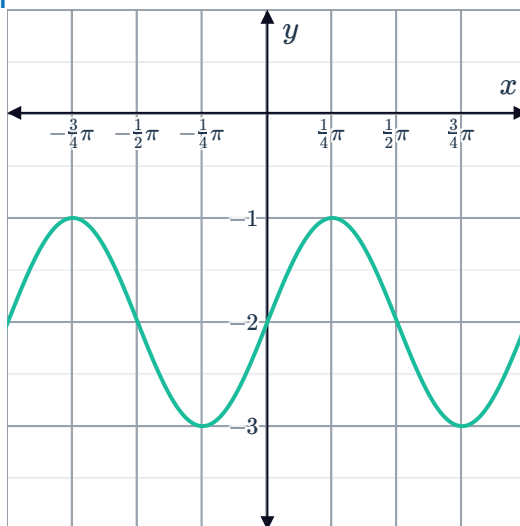
27

a

i $(-\infty, \infty)$

ii $[-3, -1]$

iii

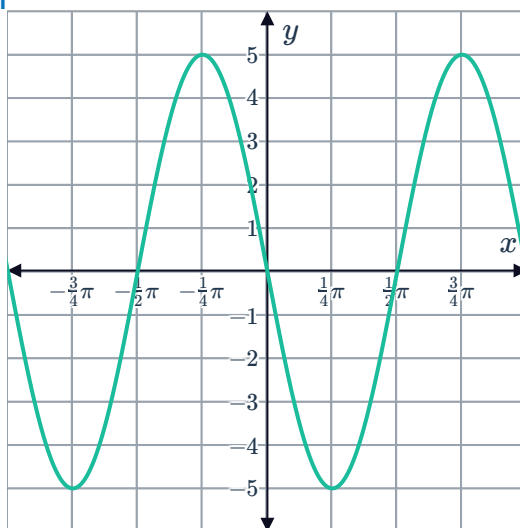


b

i $(-\infty, \infty)$

ii $[-5, 5]$

iii

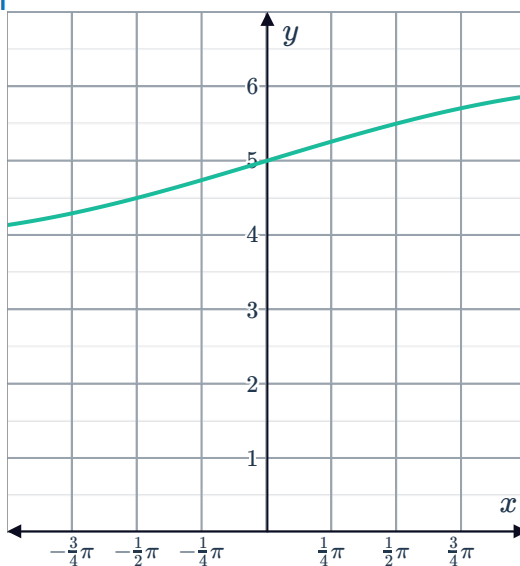


c

i $(-\infty, \infty)$

ii $[4, 6]$

iii



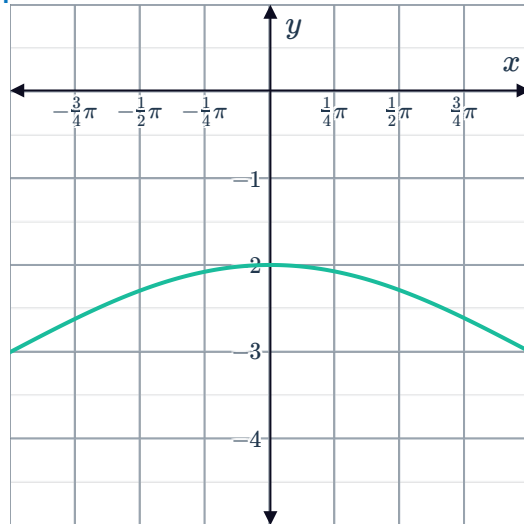
d

i $(-\infty, \infty)$

ii $[-4, -2]$

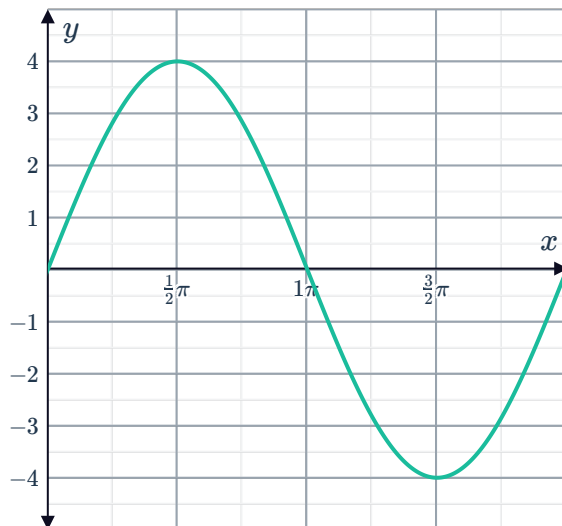


iii

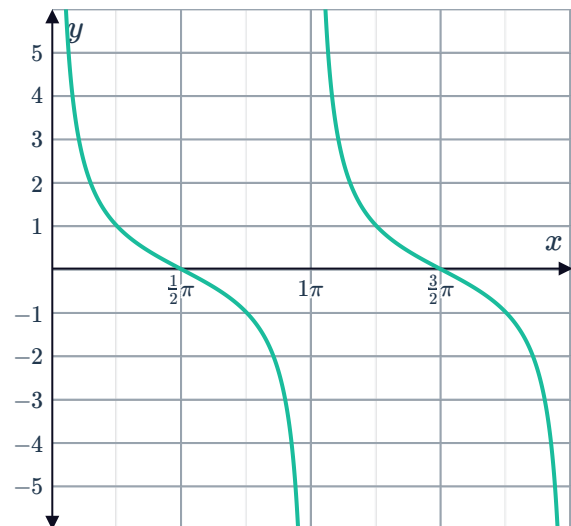


28

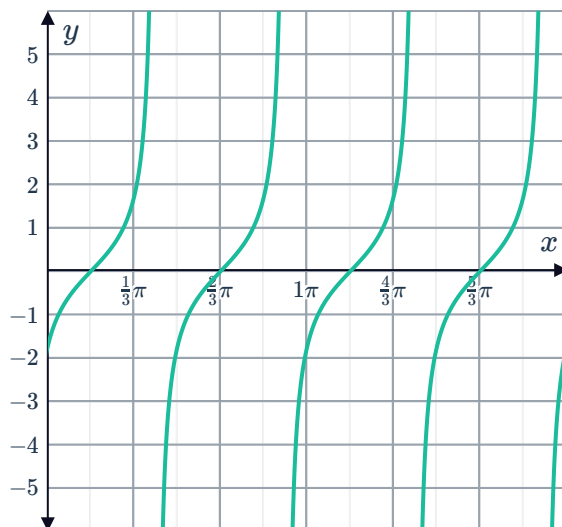
a



b



c



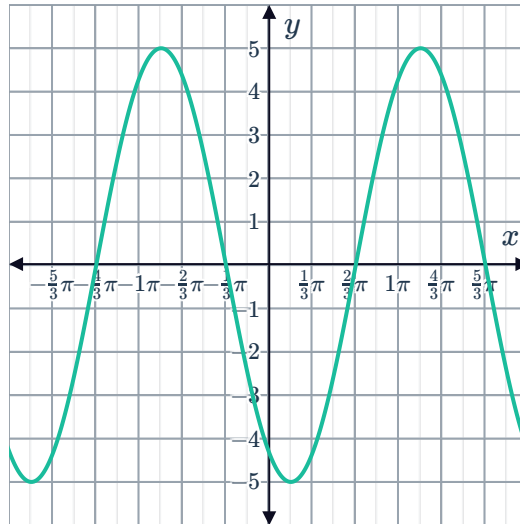
29 Dilated vertically by a factor of 2 and a phase shift of π to the right



30

- a Reflected about the x -axis, then dilated vertically by a factor of 5, followed by a phase shift of $\frac{\pi}{3}$ units to the left.

b



31 $y = -\cos\left(x + \frac{\pi}{6}\right) + 5$

32

- a $y = 3 \sin 2x$ b $y = -\sin 2x$ c $y = 2 \cos 3x$ d $y = -4 \cos\left(\frac{x}{2}\right)$

33

- a 3
b $\frac{\pi}{3}$
c Translated 3 units downwards and a phase shift of $\frac{\pi}{3}$ to the right.

34 Translated vertically by 2 units upwards and a phase shift of $\frac{\pi}{4}$ units to the right.

35

- a Yes b No c No d Yes
e No

36

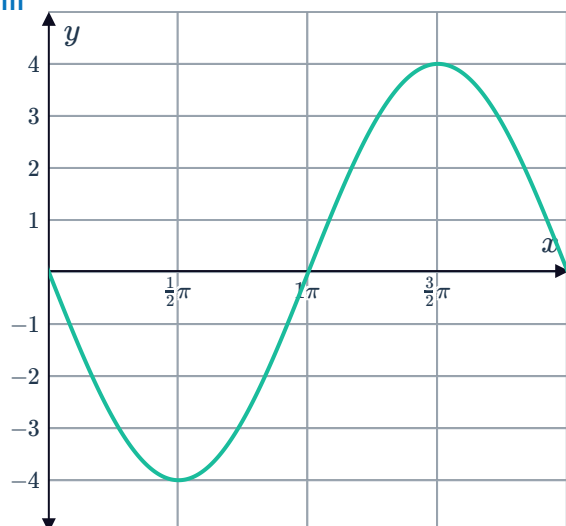


a

i 4

ii Translated π units to the right.

iii

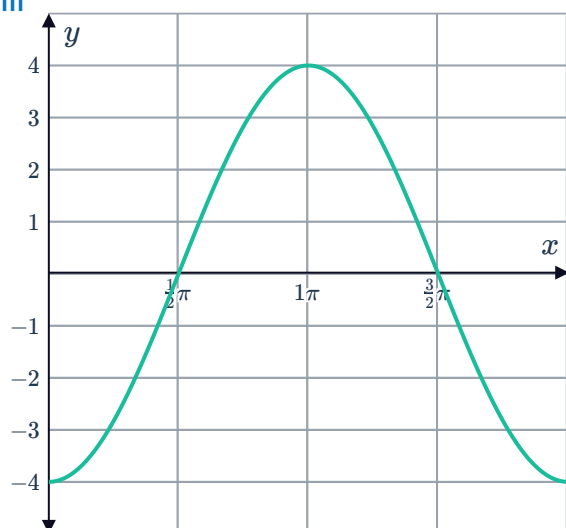


b

i 4

ii Translated $\frac{\pi}{2}$ units to the left.

iii



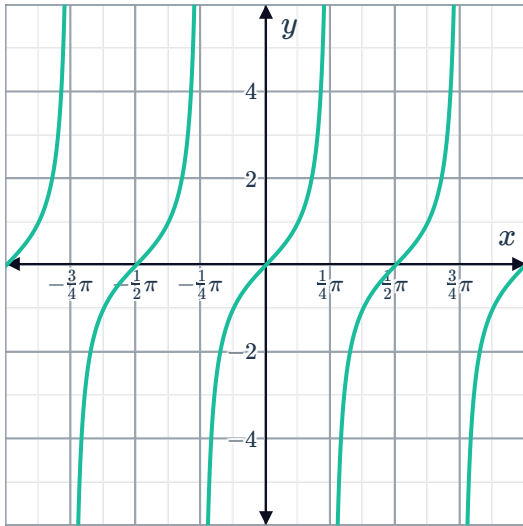
37

a

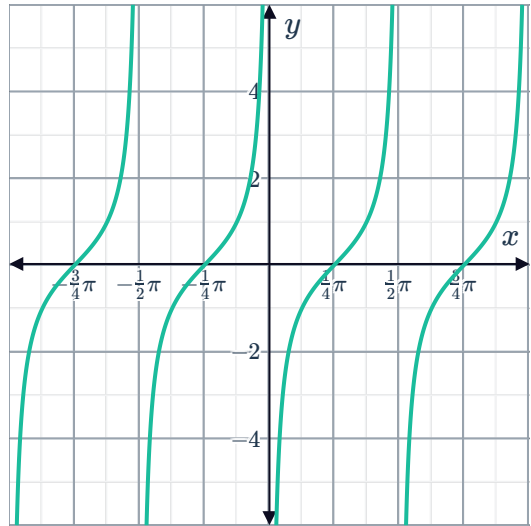
Horizontal dilation by a factor of $\frac{1}{2}$.

b

i



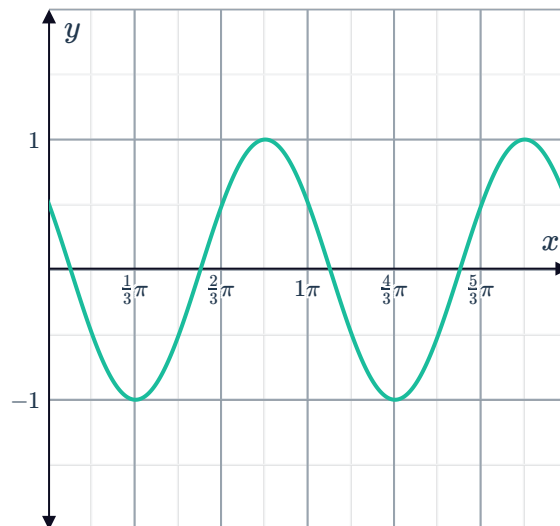
ii



38

a Horizontal dilation by a factor of $\frac{1}{2}$ and then a phase shift of $\frac{\pi}{6}$ to the left.

b



39

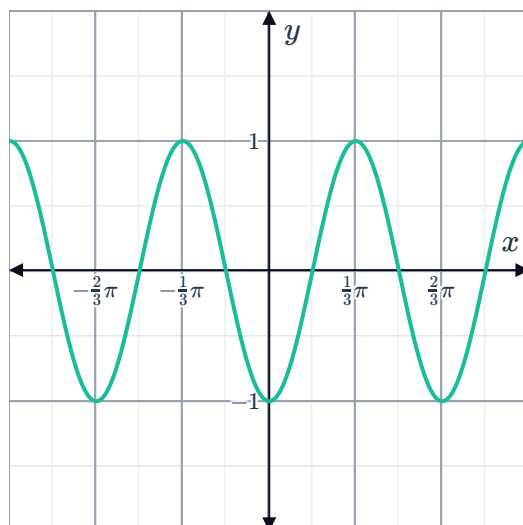
A horizontal dilation of scale factor $\frac{1}{2}$ followed by a horizontal translation of $\frac{\pi}{4}$ units to the right.

40

a $\frac{2\pi}{3}$

b Horizontal dilation by a factor of $\frac{1}{3}$ followed by a phase shift of $\frac{\pi}{3}$ units to the left.

c



41

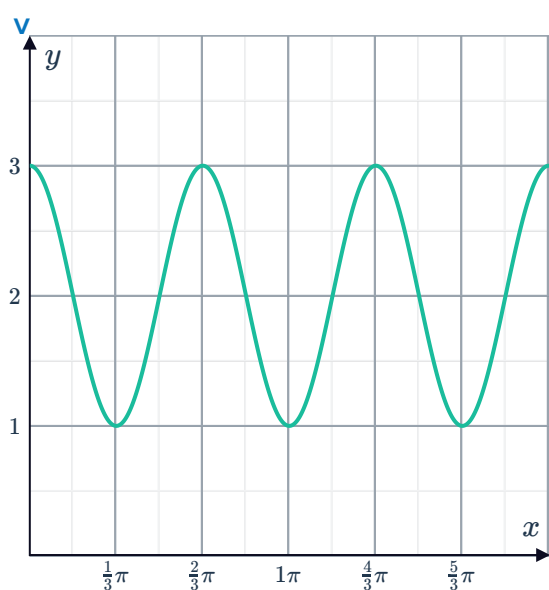
a

i $\frac{2\pi}{3}$

ii 1

iii 3

iv 1



b

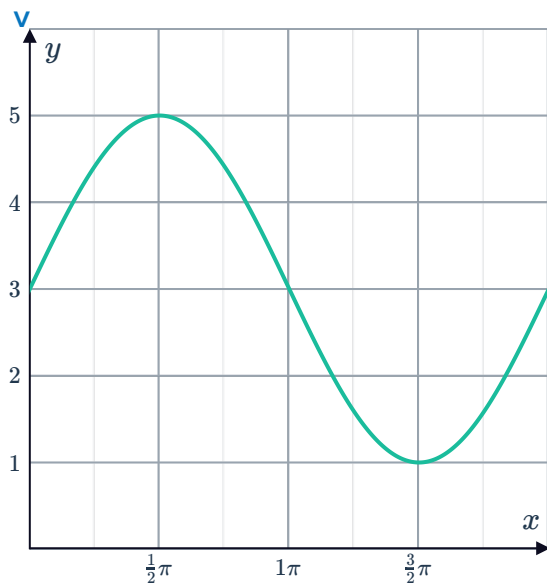
i 2π

ii 2



iii 5

iv 1



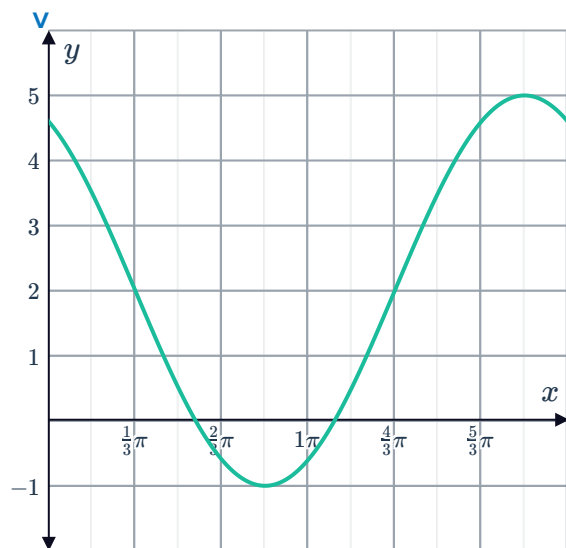
c

i 2π

ii 3

iii 5

iv -1



42



a 5π

b

- Reflected about the x -axis.
- Vertically dilated by a factor of 4.
- Horizontally dilated by a factor of 5.
- Horizontally translated by $\frac{\pi}{4}$ units to the left.

c $(-\infty, \infty)$